

Secondary Examination, 2019

Mathematics

Time- $3\frac{1}{4}$ Hours

M.M- 70

GENERAL INSTRUCTIONS TO THE EXAMINEES:

- 1) There are in all seven questions in this question paper.
- 2) All the questions are compulsory.
- 3) In the beginning of each, the number of parts to be attempted has been clearly mentioned.
- 4) Marks allotted to the questions are indicated against them.
- 5) Start from the first question and proceed to the last one.
- 6) Do not waste your time over a question you cannot solve

1. a) Find the distance between the points are (5, 0) and (-12, 0).

Sol.

We know that,

$$\begin{aligned} d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(-12 - 5)^2 + (0 - 0)^2} \\ &= \sqrt{289} = 17 \end{aligned}$$

Hence, the value is 17 .

- b) The common difference of AP is -4 and its 10th terms is -8. Determine the first terms of AP.

Sol.

Simplify the expression.

We know that,

$$T_n = a + (n - 1)d$$

$$T_{10} = 0 + (10 - 1) \times (-4)$$

$$-8 = a - 36$$

$$-8 + 36 = a$$

$$a = 28$$

Hence, the value of $a = 28$.

c)) which of the following numbers cannot be an condition.

(i) $\frac{1}{3}$ (ii) $\frac{2}{3}$ (iii) $\frac{3}{2}$ (iv) 5%

Sol.

Simplify the expression,

The given value of number cannot be an condition that is $\frac{3}{2}$.

Hence, the value of $\frac{3}{2}$.

d) Evaluate $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ}$.

Sol.

Simplify the expression,

$$\begin{aligned} & \frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} \\ & \frac{2 \times \frac{1}{\sqrt{3}}}{1 - \left(\frac{1}{\sqrt{3}}\right)^2} = \frac{2}{\sqrt{3}} \div \frac{2}{3} = \frac{2 \times 3}{\sqrt{3} \times 2} \\ & = \sqrt{3} \end{aligned}$$

Hence, the value of $\sqrt{3}$.

e) Find the value of $\angle B$, if the $\triangle ABC$ in $AB = 6\sqrt{3}cm$, $AC = 12cm$ and $BC = 6cm$.

Sol.

Simplify the expression,

We use the Pythagoras theorem.

$$AC^2 = 12^2 = 144$$

$$AC^2 + BC^2 = (6\sqrt{3})^2 + 6^2$$

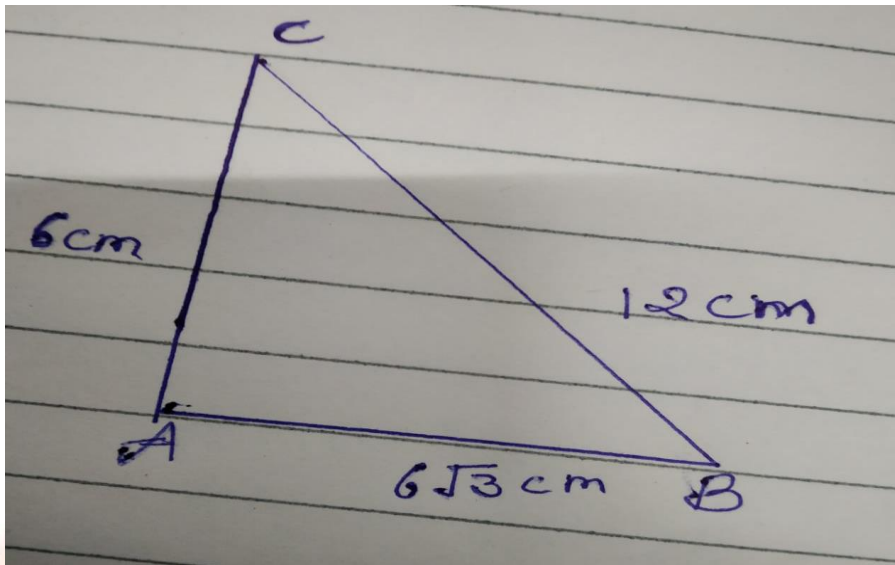
$$= 36 \times 3 + 36$$

$$= 108 + 36$$

$$= 144$$

$$= 12$$

Hence, the value of 12.



f) Find the value of k if the quadratic equation root is common $3x^2 - 6x + k = 0$.

Sol.

Simplify the expression.

We know that,

$$b^2 - 4ac = 0$$

$$D = 0$$

$$= (-6)^2 - 4 \times 3 \times k = 0$$

$$= 36 - 12k = 0$$

$$k = \frac{36}{12} = 3$$

$$k = 3$$

Hence, the value of $k = 3$

1. a) Determine the H.C.F Euclid's division its number of 130 and 280.

Sol.

Simplify the expression,

$$a = bq + r \quad \text{where } 0 < r < b$$

$$280 = 130 \times 2 + 20$$

$$130 = 20 \times 6 + 10$$

$$20 = 10 \times 2 + 0$$

$$H.C.F = 10$$

Hence, the value of H.C.F 10.

b) Find the mode given the distribution frequency.

Class interval	Frequency
10-20	14
20-30	13
30-40	12
40-50	20
50-60	11
60-70	15

Sol .

Simplify the expression,

We know that,

$$= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

$$40 + \frac{20 - 12}{2 \times 20 - 12 - 11} \times 10$$

$$= 40 + \frac{8}{17} \times 10$$

$$= 40 + 4.70$$

$$\text{Mode} = 44.70.$$

Hence, the value of 44.70.

c) find the value of $\sin A$.if the $8 \tan A = 15$.

Sol.

Simplify the expression,

$$8 \tan A = 15$$

$$\tan A = \frac{15}{8}$$

$$AC^2 = AB^2 + BC^2$$

$$= 8^2 + 15^2$$

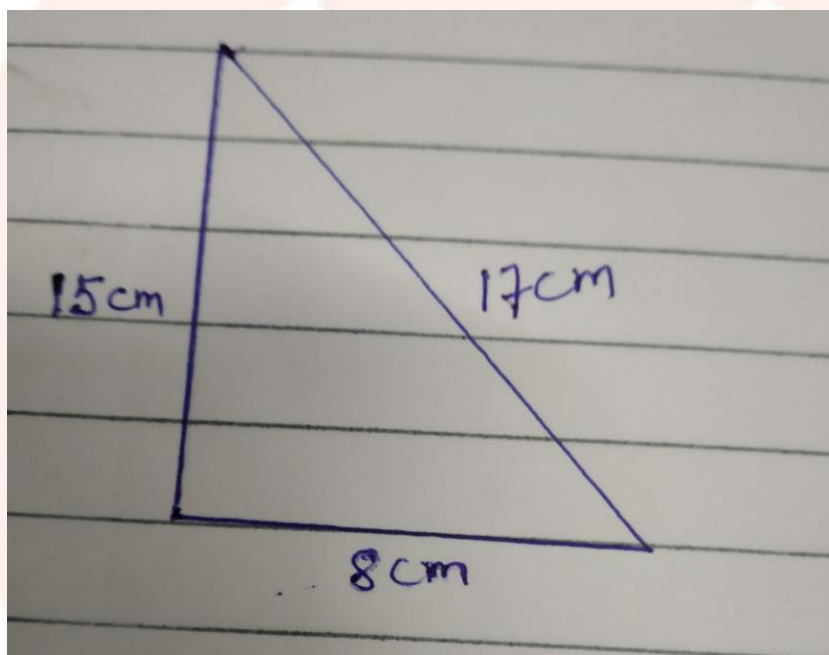
$$= 64 + 225$$

$$= 289$$

$$= 17$$

$$\text{Then, } \sin A = \frac{15}{17}.$$

$$\text{Hence, the value of } \sin A = \frac{15}{17}.$$



d) Find the area of triangle ratio . whose the euqaliteral triangle side ratio are 3:4.

Sol.

Simplify the expression,

We know that,

$$\frac{\text{first } \Delta \text{ of area}}{\text{second } \Delta \text{ of area}} = \left(\frac{\text{first } \Delta \text{ of side}}{\text{second } \Delta \text{ of side}} \right)^2$$

$$= \left(\frac{4}{9} \right)^2 = \frac{16}{81} = 16:81$$

3. a) show that the $5\sqrt{2}$ is a irrational number .

Sol.

Simplify the expression,

Let the $5\sqrt{2}$ is a rational number.

Then, $5\sqrt{2} = \frac{p}{q}$ their the p and q are equal indivisible number and $q \neq 0$.

$$\sqrt{2} = \frac{p}{5} q$$

$\frac{p}{q}$ a rational number.

$\frac{p}{5q}$ a rational number.

$\sqrt{2}$ a rational number.

We know that $\sqrt{2}$ is a irrational number but we examine after that $\sqrt{2}$ is a irrational number.

Hence, the value of $\sqrt{2}$ is a irrational number.

b) if A (1, 2), (4, y), (x, 6), (3, 5) be four point head of parallelogram . find the value of DC and y.

Sol.

Simplify the expression,

$$\begin{aligned} \text{AB of mid-point of O} &= \left[\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right] \\ &= \left[\frac{3+4}{2}, \frac{5+y}{2} \right] \\ &= \left[\frac{7}{2}, \frac{5+y}{2} \right] \end{aligned}$$

$$\text{AC of mid-point O} = \left[\frac{1+x}{2}, \frac{2+6}{2} \right] = \left[\frac{1+x}{2}, 4 \right]$$

Comparing by,

$$\frac{7}{2} = \frac{1+x}{2}$$

$$x = 5$$

$$\frac{5+y}{2} = 4$$

$$5+y = 8$$

$$y = 3$$

c) A box contain 90 disc. Their 1 to 90 terms are point out. One disc taken out. So point out the disc on number of probability.

(i) Two number of one term.

(ii) 5 divided of one term.

sol.

Simplify the expression,

$$A = [1, 2, 3, 4, \dots, 10, 11, 12, \dots, 89, 90]$$

Total type of = 90

$$\begin{aligned} \text{Two number of terms} &= 10, 11, \dots, 90 \\ &= 81 \end{aligned}$$

$$\begin{aligned} \text{Divided by 5 on terms} &= 5, 10, 15, \dots, 90 \\ &= 18 \end{aligned}$$

$$\begin{aligned} \text{(i) two number terms of probability} \\ &= \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} \\ &= \frac{81}{90} = \frac{9}{10} \end{aligned}$$

$$\begin{aligned} \text{(ii) 5 divided by a term of probability} &= \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} \\ \frac{18}{90} &= \frac{1}{5} \end{aligned}$$

Hence, the value of $\frac{9}{10}$ and $\frac{1}{5}$.

d) ΔABC prove that $\sec\left(\frac{B+C}{2}\right) = \operatorname{cosec}\frac{A}{2}$

Sol

$$\text{L.H.S} = \sec\left(\frac{B+C}{2}\right)$$

We know that,

$$A + B + C = 180$$

$$B + C = 180 - A$$

$$\frac{B+C}{2} = \left(90 - \frac{A}{2}\right)$$

$$= \sec\left(90 - \frac{A}{2}\right)$$

$$= \operatorname{cosec}\frac{A}{2}$$

Hence, the proved it.

4 a) prove that $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta) = \frac{1}{\tan\theta + \cot\theta}$

Sol.

Prove the expression,

Taking LHS

$$\begin{aligned}
 &= \left(\operatorname{cosec} \theta - \frac{1}{\operatorname{cosec} \theta} \right) \left(\sec \theta - \frac{1}{\sec \theta} \right) \\
 &= \left(\frac{\operatorname{cosec}^2 \theta - 1}{\operatorname{cosec} \theta} \right) \left(\frac{\sec^2 \theta - 1}{\sec \theta} \right) \\
 &= \left(\frac{\cot^2 \theta}{\operatorname{cosec} \theta} \right) \left(\frac{\tan^2 \theta}{\sec \theta} \right) \\
 &= \cot^2 \theta \cdot \sin \theta \times \tan^2 \theta \cdot \cos \theta \\
 &= \frac{\cos^2 \theta}{\sin^2 \theta} \cdot \sin \theta \times \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos \theta \\
 &= \frac{\cos^2 \theta}{\sin \theta} \times \frac{\sin^2 \theta}{\cos \theta} \\
 &= \cos \theta \cdot \sin \theta
 \end{aligned}$$

Now take RHS

$$\begin{aligned}
 &\frac{1}{\tan \theta + \cot \theta} \\
 &= \frac{1}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}} \\
 &= \frac{1}{\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \cdot \sin \theta}} \\
 &= \cos \theta \cdot \sin \theta
 \end{aligned}$$

Therefore,

LHS=RHS

(b). Find the value of k if the collinear point of (8, 1), (k, -4), and (2, -5) .

Sol.

Simplify the expression,

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$8(-4 + 5) + k(-5 - 1) + 2(1 + 4) = 0$$

$$8 \times 1 + k \times (-6) + 2 \times 5 = 0$$

$$8 - 6k + 10 = 0$$

$$-6k = 1 - 8$$

$$k = \frac{18}{6} = 3$$

$$k = 3$$

Hence, the value of $k = 3$.

c) Find the value pair of linear equation is $2x - 3 = 8$, $4x - 6y = 9$. Its value consistent or inconsistent.

sol.

$$a_2x + b_2y = c_2$$

$$\frac{a_1}{a_2} = \frac{1}{2}, \quad \frac{b_1}{b_2} = \frac{-3}{-6} = \frac{c_1}{c_2} = \frac{8}{9}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{1}{2} = \frac{1}{2} \neq \frac{8}{9}$$

Hence, the value of $\frac{1}{2} \neq \frac{8}{9}$.

d) Find the median of given the salary.

Class Interval	Frequency	Cumulative frequency
0-10	5	5
10-20	8	13
20-30	20	33
30-40	15	48
40-50	7	55
50-60	5	60

Sol.

Simplify the expression,

We know that,

$$M_e = l + \left\{ h \times \frac{\left(\frac{N}{2} - c \right)}{f} \right\}$$

$$N = 30 \quad \text{Therefore} = \frac{60}{2} = 30$$

$$c = 13$$

$$f = 20$$

$$h = 10$$

$$\begin{aligned} & 20 + \frac{30 - 13}{20} \times 10 \\ &= 20 + \frac{17}{20} \times 10 \\ &= 20 + 8.5 \\ &= 28.5 \end{aligned}$$

Hence, the value of median is 28.5.

5.

a) Solve the following pair of linear equation.

$$\frac{5}{x+y} + \frac{1}{x-y} = 2$$

$$\frac{15}{x+y} - \frac{5}{x-y} = -2$$

Sol.

Simplify the expression,

$$\frac{5}{x+y} + \frac{1}{x-y} = 2 \dots\dots\dots(i)$$

$$\frac{15}{x+y} - \frac{5}{x-y} = -2 \dots\dots\dots(ii)$$

$$\text{Let, } \frac{1}{x+y} = u, \quad \frac{1}{x-y} = v$$

$$5u + v = 2 \dots\dots(i)$$

$$15u - 5v = -2 \dots\dots(ii)$$

Equation (i) multiply by 3.

$$15u + 3v = 6$$

$$15u - 5v = -2$$

$$\Rightarrow 8v = 8$$

$$\Rightarrow v = 1$$

Equation by (i).

$$5u + 1 = 2$$

$$5u = 1$$

$$u = \frac{1}{5}$$

$$\frac{1}{x-y} = 1$$

$$x - y = 1 \dots\dots(A)$$

$$\frac{1}{x+y} = \frac{1}{5}$$

$$x + y = 5 \dots\dots(B)$$

Adding equation (A) and equation (B).

$$x - y = 1$$

$$x + y = 5$$

$$\Rightarrow 2x = 6$$

$$\Rightarrow x = 3$$

Substitute the value of x in equation (A).

$$3 - y = 1$$

$$3 - 1 = y$$

$$y = 2$$

Hence, the value of x and y are $x = 3$, $y = 2$.

b) Find the sum of between odd number 0 and 100.

Sol.

Simplify the expression,

We know that,

0 to 100 between odd numbers are 1, 3, 5, 7, 9, 99

$$a = 1, \quad d = 3 - 1 = 2, \quad l = 99$$

$$l = a + (n - 1) \cdot d$$

$$99 = 1 + (n - 1)$$

$$\Rightarrow 99 - 1 = 2n - 2$$

$$\Rightarrow 98 + 2 = 2n$$

$$\Rightarrow 2n = 100$$

$$\Rightarrow n = 50$$

Simplify further,

$$S_n = \frac{n}{2} [2a + (n - 1) \cdot d]$$

$$S_{50} = \frac{20}{2} [2 \times 1 + (50 - 1) \cdot 2]$$

$$= 25 [2 + 98] = 25 \times 100$$

$$\Rightarrow S_{50} = 2500$$

Hence, the value of $S_{50} = 2500$.

c) A solid cylinder of diameter 4cm and height 2 cm is melted and recast into toy in the shape of a right circular cone mounted on a hemisphere. Find the volume of toy.

sol.

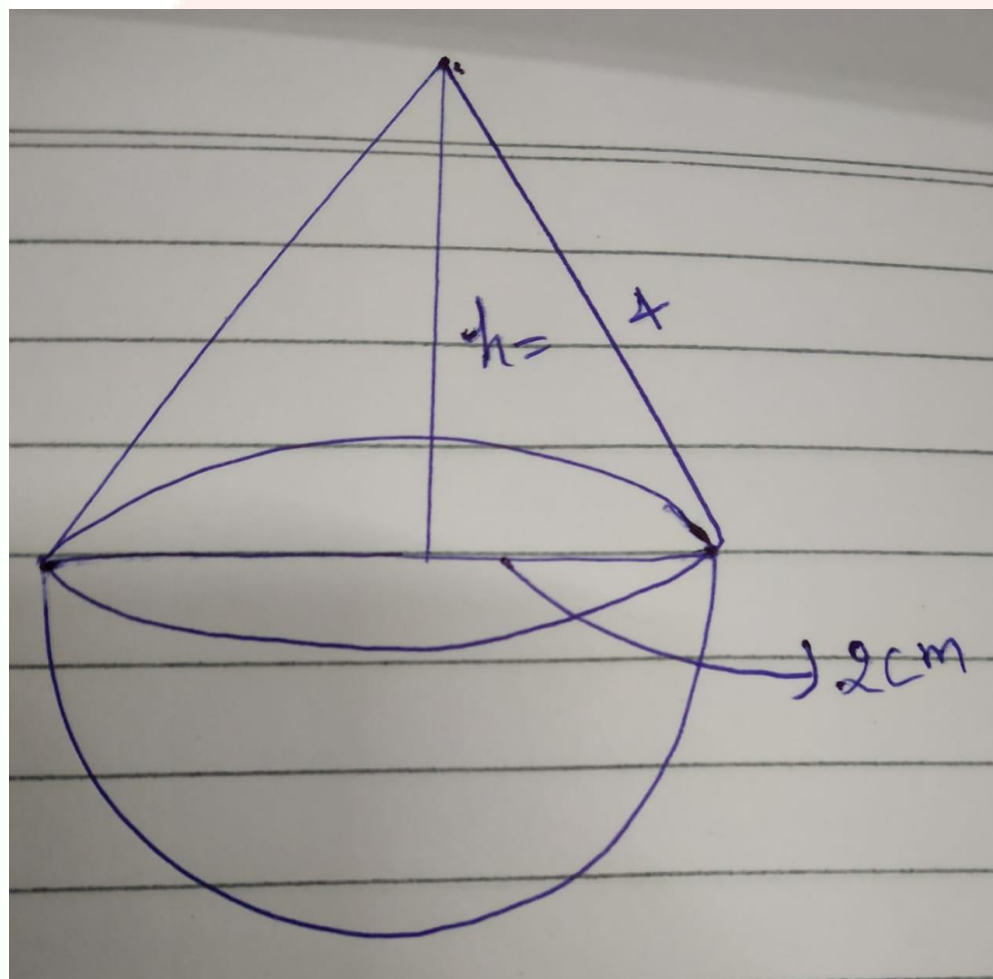
Solve the expression,

We know that,

= volume of half sphere + volume of conic.

$$\begin{aligned} & \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi r^2 (2r + h) \\ &= \frac{2}{3}\pi r^2 (2 \times 2 + 2) \\ &= \frac{1}{3}\pi \times 2^2 \times 6^2 \\ &= 8\pi = 8 \times 3.14 = 25.12 \text{ cubic cm} \end{aligned}$$

Hence, the value of 25.12 cubic cm.



d) The angle of depression from top of a light house is on a showing to ships is 30 and 45 which is 75m above the see. If the two ships are in the same side of light house and conjugative find the difference between two ship.

sol.

Simplify the expression,

Let the distance between of ships x. and $BC = y$.

Right triangle in $\triangle BCD$,

$$\tan \theta = \frac{\text{height}}{\text{base}}$$

$$\tan 45^\circ = \frac{75}{y}$$

$$1 = \frac{75}{y}$$

$$y = 75$$

Right triangle in $\triangle ACD$

$$\tan 30^\circ = \frac{DC}{AC}$$

$$\frac{1}{\sqrt{3}} = \frac{75}{x + y}$$

$$x + y = 75\sqrt{3}$$

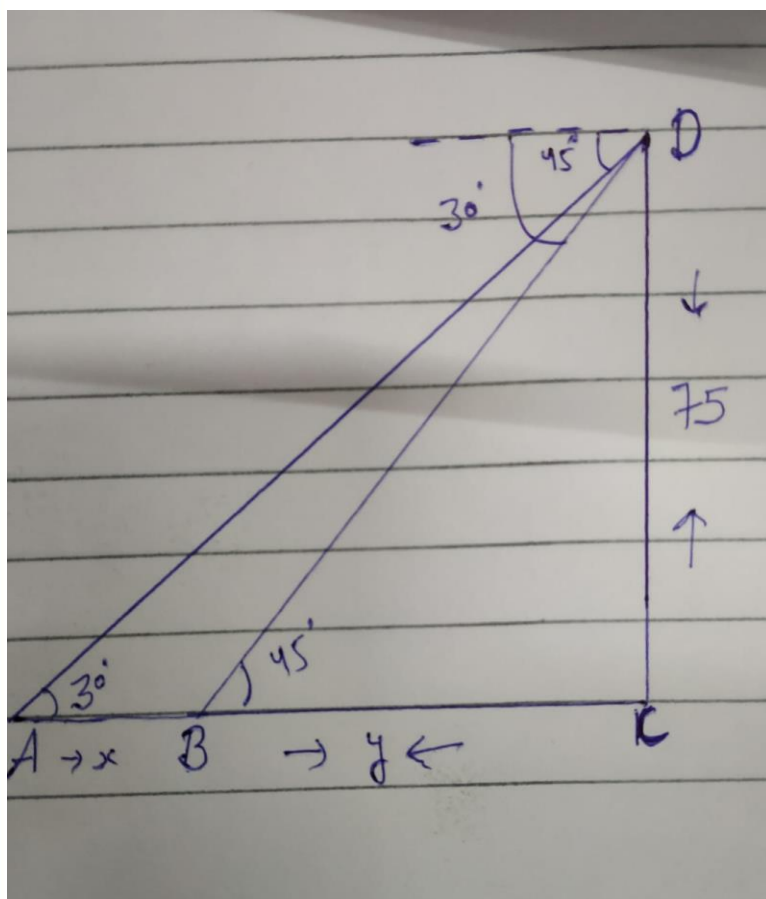
Putting the value of y.

$$x = 75(\sqrt{3} - 1)$$

$$x = 75(1.732 - 1)$$

$$x = 75 \times 0.732$$

$$x = 54.9m$$



Hence, the value of $x = 54.9m$.

6. a) Following the show compute of daily wages of a society of boys. Median of pocket allowance. Find the value of frequency F.

Daily pocket money	Number of student
11-13	7
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

Sol.

Daily pocket money	Median value	Frequency	f_x
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11-13	12	7	84
13-15	14	6	84
15-17	16	9	144
17-19	18	13	234
19-21	20	f	20f
21-23	22	5	110
23-25	24	4	96
		$\sum f_i = 44 + f$	$\sum (f_i \times x_i) = 752 + 20f$

We know that,

$$\text{Mean} = \frac{\sum (f_i \times x_i)}{\sum f_i}$$

$$18 = \frac{20f + 752}{f + 44}$$

$$18f + 792 = 20f + 752$$

$$792 - 752 = 20f - 18f$$

$$2f = 40$$

$$f = 20$$

Hence, the mean value of $f = 20$.

b) Find the area of shaded if ABCD, is region is square width is square 14cm and APD and BPC.

Sol.

Hemisphere APD of area = area of hemisphere BPC

$$= \frac{\pi}{2} r^2$$

$$= \frac{\pi}{2} \cdot \left(\frac{14}{2}\right)^2$$

$$= \frac{22}{7 \times 2} \times 49$$

$$= 77\text{cm}^2$$

Area part of the shaded = square of area – two area of hemi circle.

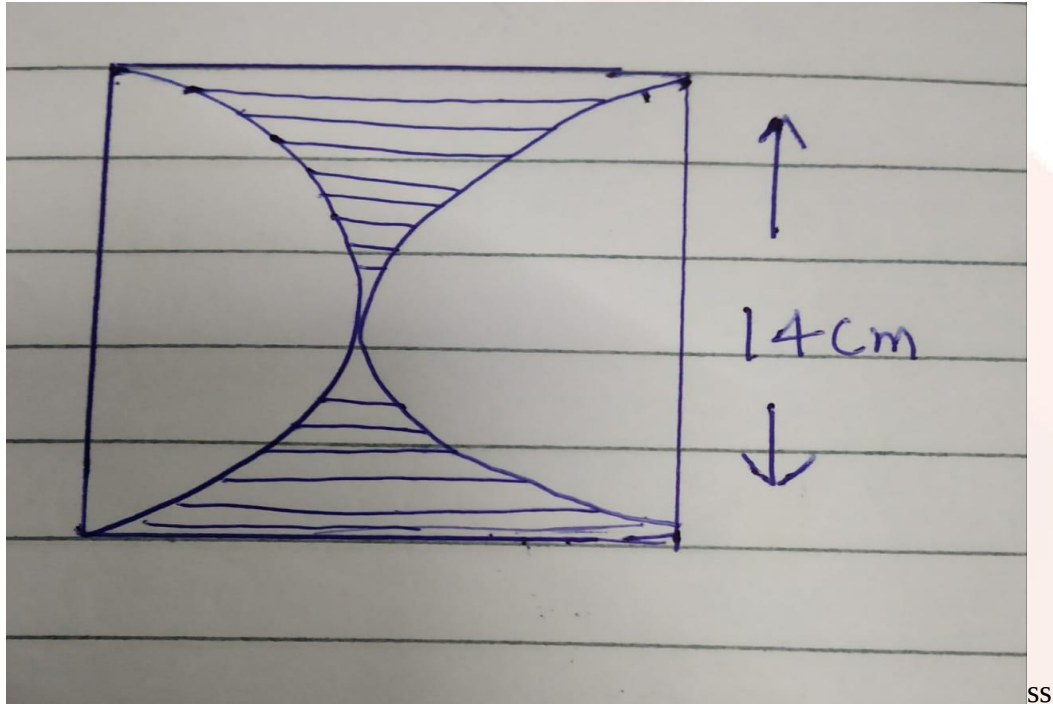
$$= \text{side}^2 - \pi r^2$$

$$= 14^2 - \frac{22}{7} \times 7^2$$

$$= 196 - 156$$

$$= 42\text{cm}^2$$

Hence, the value of 42cm^2 .



c) The lengths of tangent drawn from an external point to a circle are equal.

sol.

Simplify the expression,

Given.- two tangent AB and AC are drawn from a point A to a circle with center O.

To prove- $AC = AB$

Construction – Join OP, OQ and OA.

PROOF- AP is a tangent at P and OP is the radius through P.

$$OP \perp AP$$

Similarly, $OQ \perp AQ$

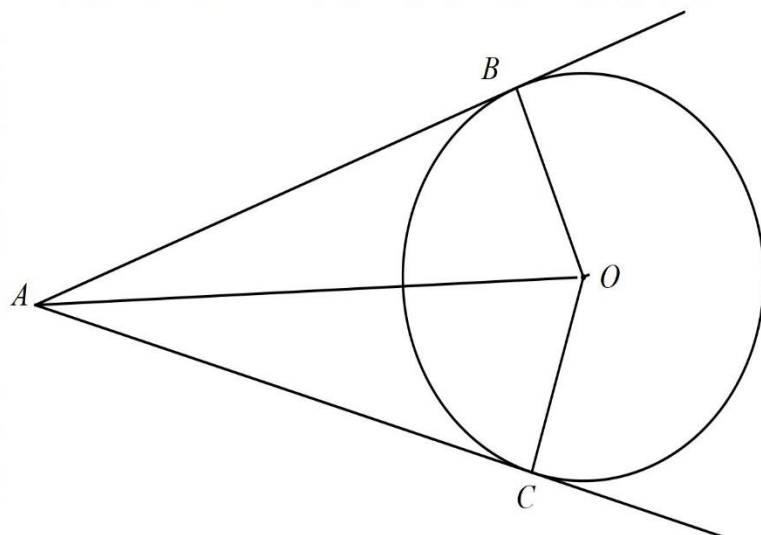
In the right angle of triangle OPA and OQA, we have:

$$OP = OQ \text{ (Radii of the same circle)}$$

$OA = OA$ (Common)

$\triangle OPA \cong \triangle OQA$ (by R.H.S)

Hence, $AP = AQ$.



7 a) If the two zeroes are $-\sqrt{\frac{5}{3}}$ and $\sqrt{\frac{5}{3}}$. Find the all zeroes value of $3x^2 + 6x^3 - 2x^2 - 10x - 5$.

Sol.

Simplify the expression,

$$x = -\sqrt{\frac{5}{3}} \quad x = \sqrt{\frac{5}{3}}$$

$$x + \sqrt{\frac{5}{3}} = 0 \quad x - \sqrt{\frac{5}{3}} = 0$$

$$\left(x + \sqrt{\frac{5}{3}}\right)\left(x - \sqrt{\frac{5}{3}}\right) = 0$$

$$x^2 - \frac{5}{3}$$

$$\frac{3x^2 + 6x^2 - 2x^2 - 10x - 5}{x^2 - \frac{5}{3}} = 3x^2 + 6x + 3$$

$$= 3x^2 + 6x + 3$$

$$= 3x^2 + 3x + 3x + 3 = 0$$

$$= 3x(x+1) + 3(x+1)$$

$$= (3x+3)(x+1)$$

$$\Rightarrow x+1 = 0$$

$$x = -1$$

$$\Rightarrow (3x+3)$$

$$\Rightarrow 3x = -3$$

$$x = -1$$

Other zero value is -1 & -1 .

Hence, the value of -1 & -1

b) In an equilateral triangle ABC, D is a point on the side BC such that $BD = \frac{1}{3}BC$. Prove that .

Sol.

$\triangle ABC$ is the right angle triangle . D is the point on BC.

$$\text{So, } BD = \frac{1}{3}BC$$

To prove : $9AD^2 = 7AB^2$

Contraction - $AB \perp BC$

Equilateral triangle $\triangle ABC$

$$BE = \frac{1}{2}BC$$

$$BE = \frac{1}{2}AB \dots \dots (1)$$

Right angle triangle ,

$$AB^2 = BE^2 + AE^2$$

$$AB^2 = BE^2 + AE^2$$

$$AB^2 = \left(\frac{1}{2} AB\right)^2 \rightarrow AE^2 \dots\dots\dots eq(1)$$

$$AB^2 = \frac{1}{4} AB^2 - AE^2$$

$$AB^2 = \frac{3}{4} AB^2 \dots\dots\dots (1)$$

Right angle triangle $\triangle AED$.

$$AD^2 = AE^2 + DE^2$$

$$AE^2 = AD^2 - DB^2$$

$$= AD^2 - \left(\frac{1}{2} BC - \frac{1}{3} BC\right)^2$$

$$= AD^2 - \left(\frac{3BC - 2BC}{6}\right)^2 \dots\dots\dots eq(2)$$

$$= \frac{3}{4} AB^2 = AD^2 - \left(\frac{AB^2}{36}\right)$$

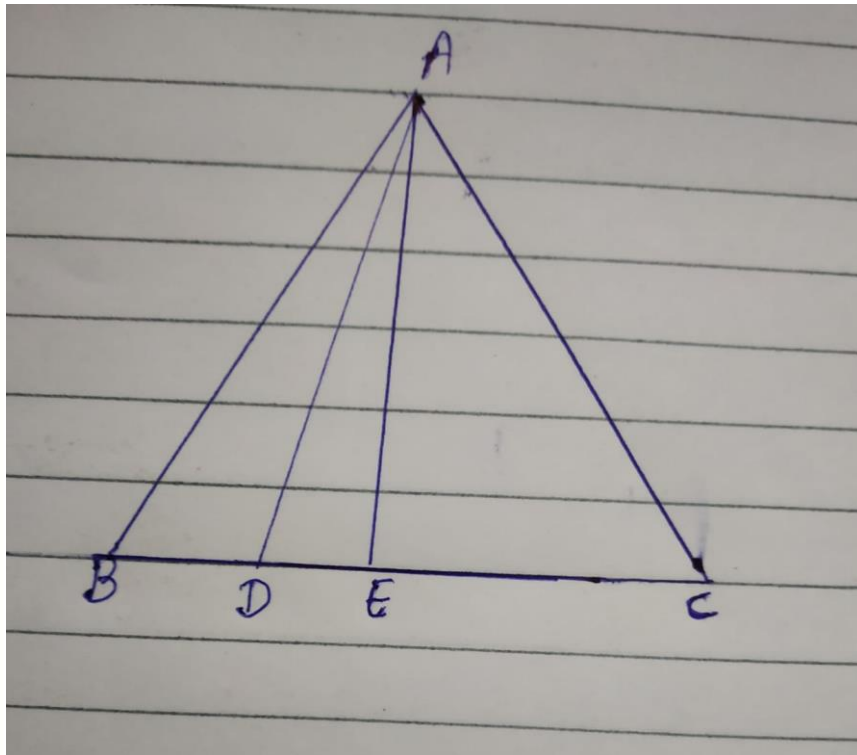
$$= \frac{3}{4} AB^2 + \frac{1}{36} AB^2 = AD^2$$

$$AD^2 = \frac{27AB^2 + AB^2}{36}$$

$$AB^2 = \frac{28AB^2}{36}$$

$$7AB^2 = 9AB^2$$

Hence, the proved it.



c) How many silver coins having diameter 1.75 and width have a melted 12cm to form a cuboid.

Sol.

Simplify the expression,

Cuboid of length = 5.5cm

Breadth = 10cm

Height 3.5cm

Volume of cuboid = $l \times b \times h$

Diameter of coin = $\frac{1.75}{2} = \frac{7}{8}$

Width = 2mm = $\frac{2}{10}$ cm = $\frac{1}{5}$ cm

Volume of coin = $\pi r^2 h$

$$= \pi \cdot \frac{7}{8} \cdot \frac{7}{8} \cdot \frac{1}{5}$$

$$= \frac{49}{320} \pi \text{ cm}^3$$

Therefore,

Volume of cuboid = volume of coin.

$$192.5 = x \frac{49\pi}{320} = n$$

$$\frac{192.5 \times 320}{49 \times \frac{22}{7}} = 400 \text{ coins}$$

Hence, the value of = 400 coins .